

A Self-Contained Proof of Graffiti.pc Conjecture 84: $t(G) \geq 2r(G)/\delta(G)$

Hai Luo

Tsinghua Shenzhen International Graduate School, Tsinghua University
Shenzhen, China

h-luo26@mails.tsinghua.edu.cn

September 21, 2026

Abstract

For a finite connected graph G , write $r(G)$ for the radius, $\delta(G)$ for the minimum degree, and $t(G)$ for the largest order of an induced subtree. Graffiti.pc Conjecture 84 (West’s catalogue, 2004) asserts

$$t(G) \geq \frac{2r(G)}{\delta(G)}.$$

Erdős, Saks, and Sós proved that every connected graph of radius r contains an induced path on at least $2r - 1$ vertices, which settles the conjecture whenever $\delta(G) \geq 2$. The remaining case is the content of this paper: the *leaf lemma*, stating that every finite connected graph with a leaf contains an induced tree on at least $2r(G)$ vertices, thereby strengthening the induced-path bound to exactly $2r$ precisely when $\delta(G) = 1$. We prove the leaf lemma and hence the conjecture. The presentation is self-contained. In particular the Erdős–Saks–Sós theorem is not used (a geodesic bound suffices for the case $\delta \geq 2$), and the radius-drop criterion together with the structure theorem for vertex-radius-decreasing graphs $G = S_\ell(B)$ —classically due to Glivak and Fajtlowicz—are re-proved here from elementary principles; a rooted form of Chung’s induced-path construction is written out in full, including two details that are usually left implicit. An accompanying Lean 4/Mathlib development verifies the leaf lemma and the final theorem under a pinned toolchain; the formalization is a supplementary verification layer.

Keywords. induced trees; graph radius; Graffiti.pc conjectures; cut vertices; radius-critical graphs; Lean 4.

1 Introduction

All graphs are finite, simple, and undirected. For a connected graph G write

$$\text{ecc}_G(v) = \max_{x \in V(G)} d_G(v, x), \quad r(G) = \min_{v \in V(G)} \text{ecc}_G(v),$$

and let $\delta(G)$ be the minimum degree. Let $t(G)$ be the maximum cardinality of a vertex set inducing a tree. A *leaf* is a vertex of degree one, $p(u)$ denotes the neighbour of a leaf u , and v is a *cut vertex* if $G - v$ is disconnected; otherwise v is a non-cut vertex.

Conjecture 1.1 (Graffiti.pc #84; West [7]). If G is connected, then $t(G) \geq 2r(G)/\delta(G)$.

For connected G with $|V(G)| \geq 2$ we have $\delta(G) \geq 1$, so the statement is equivalent to the integral inequality

$$2r(G) \leq t(G)\delta(G). \tag{1}$$

(The one-vertex graph makes $\delta = 0$; (1) then reads $0 \leq 0$ and is trivial, so we always assume $|V(G)| \geq 2$.)

The bound is tight: the path P_{2r} has $\delta = 1$ and $t = 2r$. Hence the case $\delta(G) = 1$ cannot be improved, and it is the whole difficulty:

Lemma 1.2 (Leaf lemma). *Every finite connected graph with a leaf contains an induced tree on at least $2r(G)$ vertices.*

Indeed, if $\delta(G) \geq 2$ then $t(G) \geq r(G) + 1$ by lemma 2.1 below, whence $t(G)\delta(G) \geq 2(r+1) > 2r$; if $\delta(G) = 1$ the leaf lemma gives $t(G)\delta(G) = t(G) \geq 2r$. This proves Conjecture 1.1 modulo lemma 1.2, which is proved in section 5.

The proof of the leaf lemma has two branches, according to whether deleting the leaf lowers the radius. One branch is governed by the following classical notion.

Definition 1.3. A connected nontrivial graph G is *vertex-radius-decreasing* (vrd) if $r(G - v) < r(G)$ for every non-cut vertex v of G . By lemma 2.2 the inequality is automatically an equality: $r(G - v) = r(G) - 1$.

Definition 1.4. For a connected graph B and $\ell \geq 1$, let $S_\ell(B)$ be the graph obtained from B by attaching to every vertex of B a pendant path of length ℓ (a path on ℓ new vertices, whose last vertex is a leaf of $S_\ell(B)$), using that vertex as the first endpoint.

Theorem 1.5 (Structure of vrd graphs with a cut vertex). *If G is vrd, has a cut vertex, and $r(G) \geq 2$, then $G = S_\ell(B)$ for some $\ell \geq 1$ and some connected block B (of order at least two), every vertex of which has degree $\ell + 1$ in G ; moreover*

$$r(G) = r(B) + \ell.$$

Consequently $t(G) \geq 2r(G)$ (corollary 4.9).

Classically this is due to Glivak [4] (see also the survey [5], Theorem 4, and Swart [6, Theorem 2.2.10]), with ingredients credited to Fajtlowicz [3]. Here we prove theorem 1.5 from scratch in section 4. The non-cut-vertex argument uses paths directly, without a block tree. The proof of the leaf lemma also uses a *rooted* form of Chung’s induced-path construction (lemma 3.1), proved in full in section 3, including two details that are usually left implicit.

The manuscript is organized as follows. section 2 collects the elementary facts, all proved. section 3 establishes the rooted Chung lemma, self-contained. section 4 develops theorem 1.5 and its corollary from first principles. section 5 proves the leaf lemma from the two preceding sections, and section 6 assembles Conjecture 1.1. No result of [1, 3, 4, 5, 6] is used in any proof; those works are cited only as historical credits. The companion Lean development is summarized in section 7, with a statement-by-statement correspondence in appendix A.

2 Elementary facts

Throughout, G is a finite connected graph with n vertices; v is a vertex with $G - v$ connected and nonempty when a radius $r(G - v)$ is mentioned.

Lemma 2.1 (Geodesics; the bound $t \geq r + 1$). *Every geodesic path induces a path subgraph (no chord joins two of its vertices non-consecutively). Consequently $\text{rad}(G) \leq \text{diam}(G)$ and $t(G) \geq \text{diam}(G) + 1 \geq r(G) + 1$.*

Proof. A chord would shorten the geodesic. Vertices at distance $\text{diam}(G)$ are joined by a geodesic, which is induced, of order $\text{diam}(G) + 1$; and $\text{rad}(G) \leq \text{ecc}(x) \leq \text{diam}(G)$ for every x . $\square$

Lemma 2.2 (Radius drops by at most one). *If $G - v$ is connected and nonempty then $r(G) \leq r(G - v) + 1$.*

Proof. Let c' be a centre of $G - v$ and $\rho = r(G - v)$. For $x \neq v$, $d_G(c', x) \leq d_{G-v}(c', x) \leq \rho$. The vertex v has a neighbour w in $G - v$, so $d_G(c', v) \leq 1 + \rho$. Hence $\text{ecc}_G(c') \leq \rho + 1$. $\square$

Lemma 2.3 (Leaves). *Let z be a leaf of G with $|V(G)| \geq 3$.*

- (i) *Every path with both ends $\neq z$ avoids z ; hence distances between vertices of $G - z$ agree in G and $G - z$.*
- (ii) *$r(G - z) \leq r(G)$: a centre c of G is not a leaf, and $\text{ecc}_{G-z}(c) = \text{ecc}_G(c) = r(G)$ by (i).*
- (iii) *z is not central: $\text{ecc}(z) = 1 + \text{ecc}(p(z)) \geq 1 + r(G) > r(G)$.*
- (iv) *$p(z)$ is a cut vertex (for $|V(G)| \geq 3$).*

Proof. (i) A path through z must enter and leave z along distinct edges, but z has degree one. (ii) $z \neq c$ by (iii). (iii) Clear. (iv) z is isolated in $G - p(z)$, while $G - p(z)$ has other vertices. $\square$

Lemma 2.4 (Unique eccentric points, I: the drop). *Call x the unique eccentric point (UEP) of c if $d(c, x) = \text{ecc}(c)$ and $d(c, y) < \text{ecc}(c)$ for all $y \neq x$. If c is central and has UEP v , then $G - v$ is connected and $r(G - v) = r(G) - 1$.*

Proof. Write $r = r(G)$. For $y \neq v$, a c - y path through v has length at least $d(c, v) + 1 \geq r + 1 > r - 1 \geq d(c, y)$; so every c - y geodesic avoids v , and $d_{G-v}(c, y) = d_G(c, y) \leq r - 1$. Thus every vertex of $G - v$ reaches c : $G - v$ is connected, and $\text{ecc}_{G-v}(c) \leq r - 1$. The reverse inequality is lemma 2.2. $\square$

Lemma 2.5 (Unique eccentric points, II: the converse). *If v is a non-cut vertex and $r(G - v) = r(G) - 1$, then some central vertex c of G has UEP v .*

Proof. Let c be a centre of $G - v$; then $d_G(c, x) \leq d_{G-v}(c, x) \leq r - 1$ for $x \neq v$, while $\text{ecc}_G(c) \geq r(G) = r$ forces $d_G(c, v) \geq r$. Also v has a neighbour w in $G - v$, giving $d_G(c, v) \leq 1 + d_G(c, w) \leq r$. Hence $d_G(c, v) = r = \text{ecc}_G(c)$: c is central and v is its UEP. $\square$

Lemma 2.6 (A vertex has at most one UEP). *If $x \neq x'$ are both UEPs of c , then $d(c, x') < \text{ecc}(c)$ (from x being the UEP) contradicts $d(c, x') = \text{ecc}(c)$ (from x' being the UEP).*

Lemma 2.7 (Every connected graph has at least two non-cut vertices). *The first and last vertices of a longest path are non-cut.*

Proof. Let $v_1 \dots v_k$ be a longest path and $v = v_1$. Every neighbour of v lies on it, else the path could be extended. For $x \neq v$, follow a geodesic $x = x_0, \dots, x_m = v$; the predecessor x_{m-1} is a neighbour of v , hence equals some v_i with $i \geq 2$. Then $x_0, \dots, x_{m-1}, v_i, v_{i-1}, \dots, v_2$ is a walk in $G - v$ from x to v_2 . So $G - v$ is connected. $\square$

Lemma 2.8. *If $r(G) \geq 2$ then $|V(G)| \geq 4$.*

Proof. $r \geq 2$ means no universal vertex; graphs on at most three vertices all have a universal vertex (on three vertices, both P_3 and K_3 do). $\square$

Lemma 2.9 (Counting; heredity). (i) *For $s, \ell \geq 1$: $(s + 1)(\ell + 1) - 2(s + \ell) = (s - 1)(\ell - 1) \geq 0$.*
(ii) *$t(G - v) \leq t(G)$.*

Lemma 2.10 (Auxiliary lemma). *Let D be a finite simple graph with minimum degree at least one, and let $a \in V(D)$. If every vertex $v \neq a$ has a neighbour of degree one, then a also has a neighbour of degree one.*

Proof. Suppose a has no neighbour of degree one, and let $x \in N(a)$; then $\deg(x) \geq 2$. Since $x \neq a$, x has a neighbour y of degree one, so $N(y) = \{x\}$. If $y \neq a$, applying the hypothesis to y forces $\deg(x) = 1$, a contradiction; hence $y = a$ and $\deg(a) = 1$. Pick $z \in N(x) \setminus \{a\}$, and let w be a neighbour of z of degree one ($z \neq a$). Then $w \neq a$ (as $N(a) = \{x\} \not\ni z$) and $w \neq x$ (as $\deg(x) \geq 2$), so $N(w) = \{z\}$; applying the hypothesis to w forces $\deg(z) = 1$, contradicting z being adjacent to the two distinct vertices x and w . $\square$

3 The rooted Chung lemma

Lemma 3.1 (Rooted Chung lemma). *Let F be connected with $r(F) = r \geq 2$, and let a be a non-cut vertex with $r(F - a) = r - 1$. Then F contains an induced path on at least $2r - 1$ vertices that contains a .*

Proof. Let v_0 be a centre of $F - a$, so $\text{ecc}_{F-a}(v_0) = r - 1$ and $v_0 \neq a$. Every $x \neq a$ satisfies $d_F(v_0, x) \leq d_{F-a}(v_0, x) \leq r - 1$.

Claim 0: $d_F(v_0, a) = r$. Since $\text{ecc}_F(v_0) \geq r(F) = r$ while all other vertices are at distance at most $r - 1$, we get $d_F(v_0, a) \geq r$. Conversely a has a neighbour $w \neq a$ (as $|V(F)| \geq 2$ and F is connected), and $d_F(v_0, a) \leq 1 + d_F(v_0, w) \leq r$. (This uses that a is non-isolated; the inequality $\leq r$ is the first of two details that are easy to omit.)

Fix a geodesic $v_0, v_1, \dots, v_r = a$; it is induced (lemma 2.1) and $d(v_0, v_j) = j$.

Since $\text{ecc}_F(v_2) \geq r$, there is a vertex w with $d_F(v_2, w) \geq r$; then $w \neq a$ because $d(v_2, a) = r - 2 < r$. Now

$$d_F(v_0, w) \geq d(v_2, w) - d(v_0, v_2) \geq r - 2, \quad d_F(v_0, w) \leq r - 1,$$

the latter because $w \neq a$. Let $P = p_0, \dots, p_m$ be a v_0 – w geodesic ($p_0 = v_0, p_m = w, m = d(v_0, w) \in \{r - 2, r - 1\}$); it is induced.

Chord exclusion. (1) For $j \geq 2$, no p_i is adjacent to v_j : otherwise

$$m = d(v_0, w) \geq (j - 1) + (d(v_j, w) - 1) \geq (j - 1) + (r - j + 2 - 1) = r,$$

using $d(v_j, w) \geq d(v_2, w) - d(v_2, v_j) \geq r - (j - 2)$, contradicting $m \leq r - 1$. (2) v_1 is not adjacent to any p_i with $i \geq 2$: otherwise, following v_2, v_1, p_i , then P to w , gives $d(v_2, w) \leq 1 + 1 + (m - i) \leq m \leq r - 1 < r$, a contradiction. (3) The single possible chord is $v_1 p_1$. Moreover, if $v_1 p_1$ is an edge then $m = r - 1$: the same detour as in (2) with $i = 1$ gives $r \leq d(v_2, w) \leq m + 1$, i.e. $m \geq r - 1$. (This is the second implicit detail: it is what makes the deletion below lose only one vertex.)

Conclusion. Concatenate $Q = v_r, v_{r-1}, \dots, v_0, p_1, \dots, p_m$. By (1) and (2) the only possible chord of Q is $v_1 p_1$, and $a = v_r \in Q$; the order is $(r + 1) + m$. If $v_1 p_1$ is not an edge, Q is an induced path of order at least $(r + 1) + (r - 2) = 2r - 1$. If $v_1 p_1$ is an edge, then $m = r - 1$, and deleting v_0 from Q leaves $v_r, \dots, v_1, p_1, \dots, p_m$, an induced path of order $r + (r - 1) = 2r - 1$ (now $v_1 p_1$ is a path edge and, by (2), v_1 has no other chord to P). In both cases the path contains a , which is never deleted. $\square$

Remark 3.2. The classical (unrooted) Erdős–Saks–Sós theorem [1, 2] is *not* used in this paper: the case $\delta \geq 2$ of the conjecture needs only lemma 2.1, and in the vrd branch the core block is handled with the weaker count of lemma 2.9(i).

4 The structure theorem for vrd graphs

In this section G is vrd (definition 1.3) with a cut vertex and $r := r(G) \geq 2$. Write $\text{ncv}(G)$ for the set of non-cut vertices.

Proposition 4.1 (Non-cut vertices are leaves). *Every $v \in \text{ncv}(G)$ is a leaf.*

Proof. Suppose $v \in \text{ncv}(G)$ with $\deg(v) \geq 2$. Let

$$S = \{x : x \text{ is reachable from } v \text{ by a path whose interior and endpoints all lie in } \text{ncv}(G)\},$$

the component of v in $G[\text{ncv}(G)]$ (S contains v and is connected by construction). We claim that S is closed under neighbours. Let $w \in S$. If $\deg(w) \geq 2$, then by [lemmas 2.4 and 2.5](#) (with $w \in \text{ncv}$ and $r(G-w) = r-1$) w is the UEP of a central vertex; then every neighbour x of w satisfies $d(c, x) = r-1$, and any c - q path through x has length at least $(r-1) + 1 = r > d(c, q)$, so geodesics from c avoid x ; w has a second neighbour $y \neq x$, which reaches c avoiding x , so $G-x$ is connected: $x \in \text{ncv}(G)$. Being adjacent to $w \in S$, we get $x \in S$. If $\deg(w) = 1$, then $w \neq v$ (as $\deg(v) \geq 2$), and the ncv -path from v to w has a last edge into w ; the unique neighbour of w is the penultimate vertex of that path, which lies in S . So no edge leaves S . But S is nonempty ($v \in S$), proper (it contains only non-cut vertices, while G has a cut vertex), connected, and G is connected — a contradiction. $\square$

Corollary 4.2. *The graph G has a leaf: $\text{ncv}(G)$ consists of leaves, and $|\text{ncv}(G)| \geq 2$ by [lemma 2.7](#).*

Proof. Immediate from [proposition 4.1](#) and [lemma 2.7](#). $\square$

Proposition 4.3 (Leaf stripping). *Let L be the set of leaves of G and $G_1 = G - L$. Then G_1 is connected and $r(G_1) = r - 1$. Moreover every vertex of G_1 has degree at least two in G_1 .*

Proof. G_1 is connected (deleted vertices are leaves). Lower bound: a centre c of G_1 satisfies $d_G(c, x) = d_{G_1}(c, x) \leq r(G_1)$ for non-leaves x (paths avoid leaves, [lemma 2.3\(i\)](#)) and $d_G(c, z) \leq 1 + r(G_1)$ for leaves z ; hence $r(G) \leq r(G_1) + 1$, i.e. $r(G_1) \geq r - 1$. Upper bound: $r(G - u) = r - 1$ for a leaf u ([definition 1.3](#) and [lemma 2.2](#)), and deleting the remaining leaves one by one never increases the radius ([lemma 2.3\(ii\)](#)). $\square$

Proposition 4.4 (The penultimate layer). *Let L, G_1 be as above and let w be a non-cut vertex of G_1 . Then w is a cut vertex of G and has at least one leaf neighbour.*

Proof. $w \notin L$ (leaves of G are isolated in G_1 , and $|V(G_1)| \geq 2$). If w were a non-cut vertex of G , [proposition 4.1](#) would make w a leaf, a contradiction; so w is a cut vertex of G . Now $G - w$ is disconnected, while $G_1 - w$ is connected; so $G_1 - w$ lies inside one component of $G - w$, and every other component consists of leaves only. A component of $G - w$ consisting of leaves is a single vertex adjacent to w (two leaves cannot be adjacent: each would have its neighbour outside the component). Such components exist because $G - w$ is disconnected, so w has a leaf neighbour. $\square$

Proposition 4.5 (Two leaves do not share a parent). *A vertex of G has at most one leaf neighbour.*

Proof. Suppose leaves u, u' share the parent b . By [lemma 2.5](#) (applied to the non-cut vertex u) some central c has UEP u , so $d(c, b) = r - 1$ and $c \neq b$ (else $r = d(b, u) = 1$). Every c - u' path ends with $b u'$, so $d(c, u') = d(c, b) + 1 = r$, contradicting the uniqueness in $\text{UEP}(c, u)$ (as $u' \neq u$). $\square$

Proposition 4.6 (Stripping preserves vrd). *$G_1 = G - L$ is vrd : for every non-cut vertex w of G_1 , $r(G_1 - w) = r(G_1) - 1 = r - 2$.*

Proof. Fix such w . By [proposition 4.4](#), w has a leaf neighbour u ; by [proposition 4.5](#) it is the unique one. Let c be central in G with UEP (c, u) ([lemma 2.5](#) applied to $u \in \text{ncv}(G)$): $d(c, u) = r$, all others $\leq r - 1$, so $d(c, w) = r - 1$; also $c \notin L$ ([lemma 2.3\(iii\)](#)), so $c \in V(G_1)$ and $\text{ecc}_{G_1}(c) = r - 1 = r(G_1)$.

Claim: $d(c, z) \leq r - 2$ for all $z \in V(G_1) - w$. Suppose $d(c, z) = r - 1$ (distances agree with G by [lemma 2.3\(i\)](#)). Then $z \notin L$ and $z \neq w$, so z is a cut vertex of G (otherwise [proposition 4.1](#) puts z in L). Let $C_0 \ni c$ be the component of $G - z$ and let C be another component, nonempty. For $y \in C$, every c - y path passes through z , so $d(c, y) \geq d(c, z) + 1 = r$; since $\text{ecc}(c) = r$, we get $d(c, y) = r$, hence $y = u$ by $\text{UEP}(c, u)$. So $C = \{u\}$. But w is adjacent to u and $w \neq z$, so $w \in C$, i.e. $w = u$ — absurd (as u is a leaf and w has degree at least two in G_1). This proves the claim.

Hence $\text{ecc}_{G_1-w}(c) \leq r - 2$, so $r(G_1 - w) \leq r - 2$; the reverse bound is [lemma 2.2](#) applied in G_1 . $\square$

Proposition 4.7 (Decreasing blocks). *Let B be a connected graph with no cut vertex such that $r(B - v) = r(B) - 1$ for every v (e.g. the terminal graph of the stripping chain, by [proposition 4.6](#)). Then B is self-centered and admits a fixed-point-free involution $f : V(B) \rightarrow V(B)$ with $\text{UEP}(f(v), v)$ for every v ; in particular each vertex has a unique eccentric vertex.*

Proof. For each v , [lemma 2.5](#) gives a central c_v with $\text{UEP}(c_v, v)$; by [lemma 2.6](#) the map $f : v \mapsto c_v$ is injective. Hence $|V| \leq |C(B)| \leq |V|$, so $C(B) = V(B)$: B is self-centered, and f is a bijection with $f(v) \neq v$ (as $d(v, f(v)) = r(B) \geq 1$). If $f(f(v)) \neq v$, then $f(f(v))$ and v are both at distance $r(B)$ from $f(v)$ (the former because $\text{UEP}(f^2(v), f(v))$ says $d(f^2(v), f(v)) = r(B)$), contradicting uniqueness of the eccentric vertex of $f(v)$; so $f^2 = \text{id}$. Finally $\text{UEP}(v, f(v))$ is the statement $\text{UEP}(f(f(v)), f(v))$ read at v . $\square$

Theorem 4.8 (Structure, self-contained version of [theorem 1.5](#)). *$G = S_k(B)$ for the terminal block B of the stripping chain and $k = r - r(B) \geq 1$; in particular $r(G) = r(B) + k$.*

Proof. Form the stripping chain $G = G_0 \supset G_1 \supset \dots$, $G_{i+1} = G_i - L(G_i)$ ($L(\cdot)$ the leaf set). By [propositions 4.3](#) and [4.6](#), every G_i with a cut vertex is vrd and $r(G_i) = r - i$; since $|V(G_i)|$ strictly decreases, the chain terminates at the first k with $G_k =: B$ having no cut vertex, and $r(B) = r - k \geq 1$, so $|V(B)| \geq 2$.

We show by downward induction on i that $G_i = S_{k-i}(B)$. For $i = k$ this holds with $S_0(B) := B$. Assume $G_{i+1} = S_m(B)$ with $m = k - i - 1 \geq 0$, and let P be the set of parents of $L(G_i)$; each leaf of G_i has its parent in G_{i+1} (a leaf whose parent were also deleted would force $G_i = K_2$, whose radius 1 contradicts $r(G_i) = r - i \geq r - k + 1 \geq 2$), and each vertex has at most one leaf ([proposition 4.5](#)). We must show that P is exactly the set of the $|B|$ tips of $G_{i+1} = S_m(B)$ (the vertices at depth m , one on each pendant path).

(a) *Every tip is in P .* Since $r(G_i) = r(B) + m + 1$, every $z \in B$ satisfies $\text{ecc}_{G_i}(z) \geq r(B) + m + 1$. Within $G_{i+1} = S_m(B)$, a vertex at depth j on the branch over b' is at distance at most $d(z, b') + j \leq r(B) + m$ from z (as $d(z, b') \leq r(B)$, B being self-centered). So some $p \in P$ has $d(z, p) \geq r(B) + m$; also $d(z, p) \leq r(B) + \text{depth}(p) \leq r(B) + m$. Hence $d(z, p) = r(B) + m$ and the witness is unique: B has a unique eccentric vertex $f(z)$ ([proposition 4.7](#)), and the distance bound shows p lies on $f(z)$'s branch at depth m , i.e. p is $f(z)$'s tip. As f is surjective, every tip is in P .

(b) *No other vertex is in P .* Suppose $p \in P$ has depth $j < m$ on the branch over $b \in B$, with leaf u adjacent to p . The vertex u is a non-cut vertex of G_i , so by [lemma 2.5](#) some centre c of G_i has $\text{UEP}(c, u)$: $d(c, u) = r(B) + m + 1$, hence $d(c, p) = r(B) + m$. Now c is not a leaf of G_i ([lemma 2.3\(iii\)](#)). If $c \in B$, then $d(c, p) \leq r(B) + j \leq r(B) + m - 1$, too small. If c lies on p 's own branch at depth j' , then $d(c, p) = |j' - j| \leq m$, too small. If c lies on the branch over $b'' \neq b$ at depth j'' , let w be the tip of p 's branch (depth m ; it exists and lies in G_{i+1}). Then

$$d(c, w) = j'' + d(b'', b) + m = d(c, p) + (m - j) \geq r(B) + m + 1 = \text{ecc}(c),$$

so $d(c, w) = \text{ecc}(c)$, and uniqueness in $\text{UEP}(c, u)$ forces $w = u$; but $w \in V(G_{i+1})$ while $u \notin V(G_{i+1})$, a contradiction.

This completes the induction: $G = G_0 = S_k(B)$, and $r(G) = r(B) + k$ because the stripping chain drops the radius by exactly one per step ([propositions 4.3](#) and [4.6](#)). Moreover every vertex of B has degree $\ell + 1$ in G for the appropriate layer, as asserted in [theorem 1.5](#). $\square$

Corollary 4.9 ($t \geq 2r$ for vrd graphs with a cut vertex). *If G is vrd with a cut vertex, then $t(G) \geq 2r(G)$.*

Proof. Write $G = S_k(B)$ as in [theorem 4.8](#), $s = r(B) \geq 1$. Take a geodesic Q of B (order at least $s + 1$, [lemma 2.1](#)). The union of $V(Q)$ with, for each $q \in Q$, the ℓ -vertex pendant path attached to q , induces a tree: pendant paths are internally disjoint, there are no edges between distinct pendant paths, and Q is induced in B ([lemma 2.1](#)); its order is at least

$$(s + 1)(k + 1) \geq 2(s + k) = 2r(G)$$

by [lemma 2.9\(i\)](#). □

Remark 4.10. The case $B = K_2$ needs no special treatment: K_2 satisfies [proposition 4.7](#) with f the transposition, and equality holds throughout [corollary 4.9](#) ($G = P_{2k+2}$, $t = 2r$).

Remark 4.11 (Historical credit). Theorems of this shape are classical: Glivak [4] (see [5], Theorem 4, and Swart [6, Theorem 2.2.10]) describes radially critical graphs as a “decreasing” core with equilength endpaths, and the fact that non-cut vertices of vrd graphs with a cut vertex are leaves is attributed to Fajtlowicz [3]. The proofs above were reconstructed from first principles and are self-contained; we did not consult the original arguments.

5 The leaf lemma

Proof of lemma 1.2. Suppose the lemma fails and let G be a counterexample of minimum order: G connected with a leaf u and $t(G) < 2r(G)$, while every smaller connected graph with a leaf satisfies the lemma. Let $a = p(u)$ and $r = r(G)$.

If $r = 1$, the edge ua induces a tree on $2 = 2r$ vertices, a contradiction. So $r \geq 2$; by [lemma 2.8](#) $|V(G)| \geq 4$, and a is a cut vertex ([lemma 2.3\(iv\)](#)).

Claim 1. For every non-cut vertex $v \neq u$ of G : $r(G - v) = r - 1$. Indeed $v \neq a$ (as a is a cut vertex and v is not), so $G - v$ is connected and still has the leaf u . If $r(G - v) \geq r$, minimality gives $t(G - v) \geq 2r(G - v) \geq 2r$, while $t(G - v) \leq t(G)$ ([lemma 2.9\(ii\)](#)) — a contradiction. So $r(G - v) < r$, and [lemma 2.2](#) gives $r(G - v) = r - 1$.

Setup. Let $H = G - u$. By [lemma 2.3](#), distances among $V(H)$ agree in G and H , and u is not central; hence centres of G survive in H and $r(H) \leq r$. If $r(H) < r$, then u is a non-cut vertex and [lemma 2.2](#) gives $r(H) = r - 1$. So $r(H) \in \{r - 1, r\}$.

Case A ($r(H) = r - 1$). Then every non-cut vertex of G deletes the radius to $r - 1$: u by the case assumption, all others by Claim 1. So G is vrd with the cut vertex a , and [corollary 4.9](#) gives $t(G) \geq 2r$ — a contradiction. (Minimality is not needed in this case.)

Case B ($r(H) = r$).

Claim 2. H has no leaf; in particular $\delta(H) \geq 2$. Otherwise H is a smaller connected graph with a leaf and $r(H) = r$, so minimality gives $t(H) \geq 2r(H) = 2r$; but $t(G) \geq t(H)$ ([lemma 2.9\(ii\)](#)), contradiction.

Claim 3. For every non-cut vertex $v \neq a$ of H : $r(H - v) = r - 1$. $G - v$ is obtained from the connected graph $H - v$ by re-attaching the leaf u at a , hence is connected; so v is a non-cut vertex of G , and Claim 1 gives $r(G - v) = r - 1$. The graph $G - v$ has at least three vertices, so its leaf u is not central ([lemma 2.3\(iii\)](#)); hence centres of $G - v$ survive in $H - v$ and $r(H - v) \leq r - 1$. The reverse inequality is [lemma 2.2](#) (as $r(H) = r$ and v is a non-cut vertex of H).

Claim 4. H has no cut vertex. Claims 2 and 3 and [lemma 2.5](#) imply that every non-cut vertex $v \neq a$ is a central vertex’s UEP and has degree at least two. The neighbour argument in [proposition 4.1](#) therefore shows that every neighbour of v is also a non-cut vertex. Choose a non-cut vertex $v \neq a$; two distinct non-cut vertices exist by the longest-path argument. If a were a cut vertex, the non-cut property would propagate along any path starting at v : a non-cut vertex can never equal a , so the neighbour implication applies at every step. A path from v to a would give a contradiction. Thus a

is a non-cut vertex. For any $w \neq a$, choose a v - w path in the connected graph $H - a$. Propagating the same implication along this path shows that w is a non-cut vertex. Hence every vertex of H is a non-cut vertex.

By Claim 4 every vertex of H is a non-cut vertex. For each $v \neq a$, Claim 3 and lemma 2.5 provide a central c_v with $\text{UEP}(c_v, v)$; distinct v give distinct c_v (lemma 2.6). So with $n = |V(H)|$, at least $n - 1$ vertices of H are central.

Claim 5. H is self-centered. If some b were non-central, then every $y \neq b$ is central, whence $d(y, b) \leq r$, so $d(b, y) \leq r$ and $\text{ecc}(b) \leq r$; but always $\text{ecc}(b) \geq r(H) = r$. Hence $\text{ecc}(b) = r$ and b is central, a contradiction.

Form the auxiliary graph D on $V(H)$ by putting $xy \in E(D)$ iff $d_H(x, y) = r$. Since H is self-centered and finite, every vertex has a vertex at distance exactly r : $\delta(D) \geq 1$. For each $v \neq a$, c_v is adjacent to v in D and $\deg_D(c_v) = 1$ (any other w with $d(c_v, w) = r$ would violate $\text{UEP}(c_v, v)$). So every $v \neq a$ has a degree-one neighbour, and lemma 2.10 provides c adjacent to a in D with $\deg_D(c) = 1$. Thus $d_H(c, a) = r$ and a is the unique vertex at distance r from c ; by Claim 5, c is central, so $\text{UEP}(c, a)$. lemma 2.4 gives $r(H - a) = r - 1$, and a is a non-cut vertex of H (Claim 4). lemma 3.1 yields an induced path P of H with $|P| \geq 2r - 1$ and $a \in P$.

Finally, u is adjacent in G only to $a \in P$, so $G[V(P) \cup \{u\}]$ is an induced tree on at least $2r$ vertices — the final contradiction. $\square$

6 Graffiti.pc Conjecture 84

Theorem 6.1. *Let G be a finite connected graph with at least two vertices. Then*

$$2r(G) \leq t(G)\delta(G),$$

equivalently $t(G) \geq 2r(G)/\delta(G)$.

Proof. A connected nontrivial graph has $r(G) \geq 1$ and $\delta(G) \geq 1$. If $\delta(G) \geq 2$, lemma 2.1 gives $t(G) \geq r(G) + 1$, so $t(G)\delta(G) \geq 2(r(G) + 1) > 2r(G)$. If $\delta(G) = 1$, G has a leaf, and lemma 1.2 gives $t(G) \geq 2r(G)$, so $t(G)\delta(G) = t(G) \geq 2r(G)$. $\square$

Remark 6.2. The leaf lemma is genuinely about induced trees: one cannot strengthen it to an induced path on $2r$ vertices. In Case B of its proof the extra vertex is a pendant attachment to a long induced path; in Case A the tree is assembled from several pendant branches over a geodesic of the core block.

7 Supplementary Lean 4 formalization

An accompanying Lean 4/Mathlib development formalizes the leaf lemma and the final theorem together with their principal ingredients, under the pinned toolchain Lean and Mathlib 4.32.2. For Case A the formal proof deliberately takes a shorter route than this manuscript: direct induction by peeling leaves off a maximum induced tree, without the structural classification of section 4. The continuous-integration build of the repository compiles every module on a clean runner, scans the sources for proof placeholders and custom axioms, and prints the axiom dependencies of the final declarations, which are the three standard Lean axioms only. The formalization is supplementary: the mathematical proof in this article is self-contained, and appendix A provides a navigation map between the two developments.

References

- [1] P. Erdős, M. Saks, and V. T. Sós, *Maximum induced trees in graphs*, J. Combin. Theory Ser. B **41** (1986), 61–79. [https://doi.org/10.1016/0095-8956\(86\)90028-6](https://doi.org/10.1016/0095-8956(86)90028-6)

- [2] F. R. K. Chung, proof of the induced-path bound, Theorem 2.2 of [1]; lemma 3.1 is a rooted variant of her construction, with the proof written out in full.
- [3] S. Fajtlowicz, *A characterization of radius-critical graphs*, J. Graph Theory **12** (1988), 529–532. <https://doi.org/10.1002/jgt.3190120409>
- [4] F. Glivjak, *On the structure of radially critical graphs*, in *Graphs, Hypergraphs and Block Systems*, Proc. Sympos. Zielona Góra, Poland (1976), 69–73.
- [5] F. Glivjak, *On radially extremal graphs and digraphs, a survey*, Math. Bohem. **125** (2000), 215–225.
- [6] C. S. Swart, *Distance Measures in Graphs and Subgraphs*, M.Sc. thesis, University of Natal, 1996.
- [7] D. B. West, *Some conjectures of Graffiti.pc*, online catalogue, entry 84 (2004).

A Formal theorem map

This map is provided for navigation of the supplementary source and is not part of the logical dependency of the manuscript. The Case A branch of the Lean proof uses the shorter leaf-peeling induction mentioned in section 7; the remaining rows correspond to statements proved in both developments. The principal correspondences are:

Mathematical statement	Lean declaration / file
Geodesics induce paths; $t \geq r + 1$	two_radius_le_treeNumber_mul_minDegree GraphConjecture84.lean
Radius drops by at most one	radOn_le_radOn_erase_add_one BasicFacts.lean
UEP I: the drop	isUniqueEccentricPoint_of_induce_radius_drop Deletion.lean
UEP II: the converse	induce_radius_drop_of_central_uep Deletion.lean
A vertex has at most one UEP	isUniqueEccentricPoint_unique BasicFacts.lean
Two non-cut vertices	exists_two_deleteConnected EndBlocks.lean
Unique leaf neighbour	unique_leaf_neighbor EndBlocks.lean
Auxiliary degree-one lemma	degree_one_neighbor_of_all_other RadiusCriticalStructure.lean
Rooted Chung lemma	rooted_chung RootedChung.lean
Case B: non-cut closure	deleteConnected_all_of_drop_except EndBlocks.lean
Case B: all vertices central; UEP for a	isCentral_all_of_uep_except, central_uep_of_uep_except CaseB.lean
Case B: rooted induced tree	rooted_tree_of_drop_except CaseB.lean
Case A branch (peeling route)	vr_d_cut_tree_bound CaseA.lean
Leaf lemma	leafLemma LeafLemma.lean
Main theorem	graphConjecture84 GraphConjecture84.lean